Concise Data Dictionary — Review 2

Project: School Transport Route & Student Pickup Management System (Project 21)

This data dictionary defines the structure, data types, nullability, primary/foreign key relationships, and integrity constraints for all 12 relational tables normalized up to Third Normal Form (3NF). Target DBMS: PostgreSQL 15+.

1. Bus Table

Description: Master catalog of school buses in the fleet.

Attribute Name	Data Type	Key Type	Nullable	Constraints & Description
`bus_id`	`INTEGER`	**PK**	NOT NULL	Unique vehicle identifier.
`number`	`VARCHAR(20)`	**UNIQUE**	NOT NULL	Bus registration number.
`capacity`	`INTEGER`	-	NOT NULL	`CHECK (capacity > 0 AND capacity <= 60)` permitted seats, from 1 to 60.
`model`	`VARCHAR(80)`	-	NOT NULL	Vehicle model.
`status`	`VARCHAR(12)`	-	NOT NULL	`CHECK (status IN ('ACTIVE', 'SERVICE'))` operational status.

2. Driver Table

Description: Licensed drivers who operate the buses.

Attribute Name	Data Type	Key Type	Nullable	Constraints & Description
`driver_id`	`INTEGER`	**PK**	NOT NULL	Unique driver identifier.
`name`	`VARCHAR(80)`	-	NOT NULL	Driver full name.
`license_no`	`VARCHAR(15)`	**UNIQUE**	NOT NULL	`CHECK (license_no ~ '^[A-Z]{2}[0-9]{2}[0-9]{11}$')` two uppercase letters followed by 13 digits.
`phone`	`VARCHAR(15)`	-	NOT NULL	Contact number.

3. Attendant Table

Description: Bus attendants who assist students on board.

Attribute Name	Data Type	Key Type	Nullable	Constraints & Description
`attendant_id`	`INTEGER`	**PK**	NOT NULL	Unique attendant identifier.
`name`	`VARCHAR(80)`	-	NOT NULL	Attendant full name.
`phone`	`VARCHAR(15)`	-	NOT NULL	Contact number.

4. Route Table

Description: Shift-specific route assignment linking a bus, a driver, and an attendant.

Attribute Name	Data Type	Key Type	Nullable	Constraints & Description
`route_id`	`INTEGER`	**PK**	NOT NULL	Unique route identifier.
`route_name`	`VARCHAR(100)`	-	NOT NULL	Route title (e.g., 'Civil Lines to KPS Dunda').
`bus_id`	`INTEGER`	**FK**	NOT NULL	References `Bus(bus_id)`.
`driver_id`	`INTEGER`	**FK**	NOT NULL	References `Driver(driver_id)`.
`attendant_id`	`INTEGER`	**FK**	NOT NULL	References `Attendant(attendant_id)`.
`shift`	`VARCHAR(10)`	-	NOT NULL	`CHECK (shift IN ('MORNING', 'AFTERNOON'))`.

5. Stop Table

Description: Geo-coded pickup and drop points.

Attribute Name	Data Type	Key Type	Nullable	Constraints & Description
`stop_id`	`INTEGER`	**PK**	NOT NULL	Unique stop identifier.
`stop_name`	`VARCHAR(100)`	-	NOT NULL	Stop or landmark name.
`latitude`	`NUMERIC(9,6)`	-	NOT NULL	`CHECK (latitude BETWEEN -90 AND 90)`.
`longitude`	`NUMERIC(9,6)`	-	NOT NULL	`CHECK (longitude BETWEEN -180 AND 180)`.

6. RouteSequence Table

Description: Associative junction defining the ordered stops along a route.

Attribute Name	Data Type	Key Type	Nullable	Constraints & Description
`route_id`	`INTEGER`	**PK, FK**	NOT NULL	References `Route(route_id)`. Part of composite key `(route_id, stop_id)`.
`stop_id`	`INTEGER`	**PK, FK**	NOT NULL	References `Stop(stop_id)`. Part of composite key `(route_id, stop_id)`.
`seq_no`	`INTEGER`	-	NOT NULL	`CHECK (seq_no > 0)`. Unique `(route_id, seq_no)`.
`scheduled_time`	`TIME`	-	NOT NULL	Scheduled arrival time at the stop.

7. Student Table

Description: Students enrolled in the school.

Attribute Name	Data Type	Key Type	Nullable	Constraints & Description
student_id	INTEGER	**PK**	NOT NULL	Unique student identifier.
name	VARCHAR(80)	-	NOT NULL	Student full name.
class	VARCHAR(8)	-	NOT NULL	Class or grade.
section	VARCHAR(4)	-	NOT NULL	Section within the class.
guardian_name	VARCHAR(80)	-	NOT NULL	Parent or guardian name.

8. FeePlan Table

Description: Term-wise transport fee plans.

Attribute Name	Data Type	Key Type	Nullable	Constraints & Description
plan_id	INTEGER	**PK**	NOT NULL	Unique fee plan identifier.
term	VARCHAR(20)	-	NOT NULL	Billing term.
amount	NUMERIC(10,2)	-	NOT NULL	CHECK (amount > 0) plan fee in INR.
academic_year	VARCHAR(9)	-	NOT NULL	Academic year the plan applies to.

9. TransportRegistration Table

Description: Assigns a student to a route stop under a fee plan.

Attribute Name	Data Type	Key Type	Nullable	Constraints & Description
reg_id	INTEGER	**PK**	NOT NULL	Unique registration identifier.
student_id	INTEGER	**FK**	NOT NULL	References Student(student_id). Partial unique index allows one ACTIVE row per student.
route_id	INTEGER	**FK**	NOT NULL	Composite FK (route_id, stop_id) references RouteSequence(route_id, stop_id).
stop_id	INTEGER	**FK**	NOT NULL	Composite FK (route_id, stop_id) references RouteSequence(route_id, stop_id).
plan_id	INTEGER	**FK**	NOT NULL	References FeePlan(plan_id).
start_date	DATE	-	NOT NULL	Date the registration begins.
status	VARCHAR(10)	-	NOT NULL	CHECK (status IN ('ACTIVE', 'INACTIVE')).

Index: `CREATE UNIQUE INDEX one_active_assignment_per_student ON TransportRegistration(student_id) WHERE status = 'ACTIVE'.`

10. PickupLog Table

Description: Daily boarding and drop events recorded per registration.

Attribute Name	Data Type	Key Type	Nullable	Constraints & Description
log_id	INTEGER	**PK**	NOT NULL	Unique log entry identifier.
reg_id	INTEGER	**FK**	NOT NULL	References TransportRegistration(reg_id).
trip_date	DATE	-	NOT NULL	Date of the trip.
event_type	VARCHAR(8)	-	NOT NULL	CHECK (event_type IN ('PICKUP', 'DROP')).
event_time	TIME	-	NOT NULL	Time the event occurred.

Table constraint: Unique (reg_id, trip_date, event_type) — one pickup and one drop per registration per day.

11. Payment Table

Description: Receipt log of transport fee payments.

Attribute Name	Data Type	Key Type	Nullable	Constraints & Description
payment_id	INTEGER	**PK**	NOT NULL	Unique receipt identifier.
reg_id	INTEGER	**FK**	NOT NULL	References TransportRegistration(reg_id).
paid_on	DATE	-	NOT NULL	Payment date.
amount	NUMERIC(10,2)	-	NOT NULL	CHECK (amount > 0 AND amount <= 50000) single payment limit in INR.
receipt_no	VARCHAR(30)	**UNIQUE**	NOT NULL	Unique receipt reference.

12. MaintenanceRecord Table

Description: Servicing history for each bus.

Attribute Name	Data Type	Key Type	Nullable	Constraints & Description
maint_id	INTEGER	**PK**	NOT NULL	Unique maintenance job identifier.
bus_id	INTEGER	**FK**	NOT NULL	References Bus(bus_id).
service_date	DATE	-	NOT NULL	Last service date.
description	VARCHAR(200)	-	NOT NULL	Work performed.
cost	NUMERIC(10,2)	-	NOT NULL	CHECK (cost >= 0) maintenance cost in INR.
next_due	DATE	-	NOT NULL	Next service due date.